1. $T(n) = T(\frac{9n}{10}) + n$ Answer:
(a)

$$T(n) = T(\frac{9n}{10}) + n$$

$$T(\frac{9n}{10}) = T(\frac{9^2 n}{10^2}) + \frac{9n}{10}$$

$$T(\frac{9^2 n}{10^2}) = T(\frac{9^3 n}{10^3}) + \frac{9^2 n}{10^2}$$

...

$$T(\frac{9^{k-1} n}{10^{k-1}}) = T(\frac{9^k n}{10^k}) + \frac{9^{k-1} n}{10^{k-1}}$$

$$T(\frac{9^k n}{10^k}) = T(1)$$

$$\begin{array}{c} n \\ \downarrow \\ \frac{9n}{10} \\ \downarrow \\ \frac{9^2 n}{10^2} \\ \downarrow \\ \ldots \\ \downarrow \\ \frac{9^{k-1} n}{10^{k-1}} \\ \downarrow \\ 1 \end{array}$$

From the last equation, we have $\frac{9^k}{10^k} \cdot n = 1$
$n = (\frac{10}{9})^k$
$k = \frac{\log n}{\log(\frac{10}{9})}$
$k = \log_{(\frac{10}{9})} n$ Thus, the depth of the recursion is $\log_{(\frac{10}{9})} n$
The total cost of the recursion is the sum of the cost at each level for the depth of the recursion, which is $\log_{(\frac{10}{9})} n$.
Thus, the total cost is given by: $\sum_{k=0}^{\log_{(\frac{10}{9})} n} (\frac{9}{10})^k \cdot n$

$\sum_{k=0}^{\log_{(\frac{10}{9})} n} (\frac{9}{10})^k \cdot n = n \cdot \frac{1-(\frac{9}{10})^{\log_{\frac{10}{9}} n+1}}{1-\frac{9}{10}}$
$n \cdot \frac{1-(\frac{9}{10})^{\log_{\frac{10}{9}} n+1}}{1-\frac{9}{10}} = n \cdot \frac{1-\frac{1}{n}\cdot\frac{9}{10}}{\frac{1}{10}}$
$T(n) = 10n - 9$
Thus, $T(n) = \Theta(n)$■

(b)

$T(1) = 1$

$T(2) = T(\frac{18}{10}) + 2 = 3.8$

$T(3) = T(\frac{27}{10}) + 3 = 5.7$

$T(100) = T(\frac{900}{10}) + 100 = 190$

This begins to look like $2n$

Hypothesis: $T(k) \leq ck$ for all $k < n$

Substitute: $T(n) \leq c \cdot (\frac{9n}{10}) + n$

$T(n) \leq n(0.9c + 1)$

This is the desired form of $n \cdot c$

$0.9c + 1 \leq c$ for all $c \geq 10$

Thus, $T(n) \leq 10n$ for all $c \geq 10$

Therefore, $T(n) = O(n)$ ■

2. $S(n) = 9S(\frac{n}{3}) + n^2$

   Answer:

   (a)

   ...

   (b)